Delphi: from candidate laws to checked theorems

Experts know the conditions behind a law. An agent applying that law to a new setting does not reliably check them. We make the check executable, and keep proved, simulated and observed apart.

PROVED **COMPILED IN LEAN 4**

A finite Turing instance, and Price identity to a conditional Hamilton regression theorem, each under stated assumptions with rejected mutants.

SIMULATED

Nonlinear solver outcomes and the agent evaluation: **PENDING**

OBSERVED

None. We ran no experiment. The released enzyme baseline is appendix only.

The caveats are not our discovery. What is new here is only that they are checked by a program before a law is applied, and that we show our own failures.

The continuum Turing conditions hold, yet a finite domain can admit no growing mode, and not monotonically in size.

Declared model: J = [[1, -2], [2, -3]], diffusivities D = (1, 10), 1-D Neumann boundaries; det(J − qD) = (5q-1)(2q-1), so the continuum band is 1/5 < q < 1/2. Admissible modes are q = (nπ/L)².

LENGTH L / Π	1/2	3/4	1	3/2	2	5/2	3	4	5	6
linear verdict	stable	stable	stable	grows	grows	stable	grows	grows	grows	grows

PROVED **COMPILED INSTANCE**

Lean 4.35.0: L = π admits no unstable mode; L = 2π admits one (n = 1, det = -1/8). Build exit 0; 3 law and domain mutants rejected. Only these two lengths are compiled.

EXACT ARITHMETIC

The whole strip above is exact rational arithmetic, not Lean-compiled. Over the reals with Mathlib, a host probe re-run by the director shows no admissible mode lies in the band at L = π.

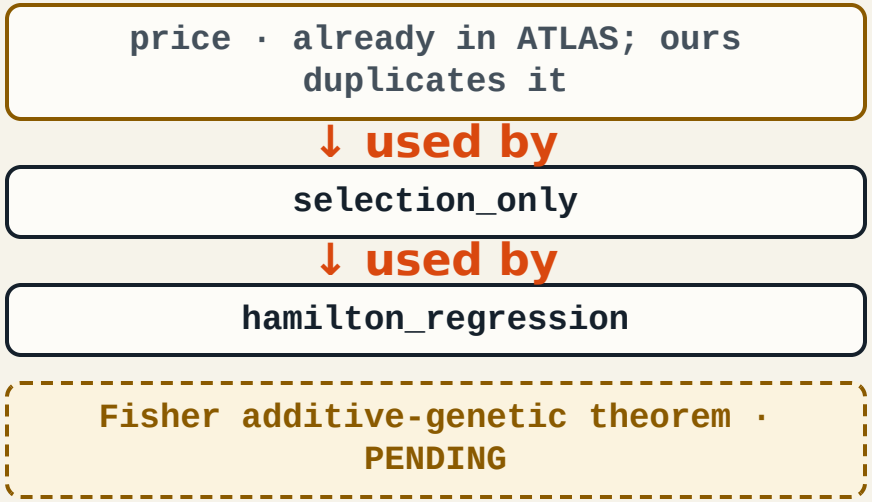
SIMULATED / **OBSERVED**

Nonlinear solver check: **PENDING**. Observed tissue pattern: **none**.

Limitation first. The algorithm is trivial: count admissible modes in the band; exact for rational L/π. The offer is a certificate and packaging, not new mathematics. Exactly, mode n grows iff $2n^2 < (L/\pi)^2 < 5n^2$: growth for L/π in (√2, √5) and every L/π above 2√2; stable window $\sqrt{5} \le L/\pi \le 2\sqrt{2}$. Non-monotone size dependence is textbook-derivable and not claimed as new; the literature citation is being verified.

Scope. A linear-mode statement for this declared model, not whether a pattern forms; we are not saying Turing was wrong.

Price identity, then a selection-only lemma, then a conditional Hamilton regression theorem: a real proof dependency.



The dependencies are read from the Lean source. A fixed-fitness variance corollary also compiles; it is **not** Fisher's theorem.

DROP AN ASSUMPTION: THE EXACT COMPILED WITNESS

ASSUMPTION DROPPED	WHAT THE WITNESS SHOWS
perfect transmission	mean fitness × trait change = 1 while selection covariance = 0
residual orthogonality	trait mean rises although r·b − c = -1
correct sign of the cost term	the sign-reversed criterion is false on the same population
premises satisfiable?	yes: change = -1/6, variance 1/4

PROVED RECEIPT

Lean exit 0; axioms: propext, Classical.choice, Quot.sound only. 4 witness controls exit 0. 2 changed-target mutants rejected (exit 1). Statement-fidelity review by a second reader: PENDING

Established mathematics. Price 1970; the Price–Fisher–Hamilton link is in Frank 1997. The Price identity is already formalized in the ATLAS biology library (facebookresearch/atlas-lean), so our version duplicates it. A critic's code search found no Hamilton form there; that search is not exhaustive. This is a statistical regression statement, not a causal cost–benefit rule.

Our own near-vacuous theorem compiled, passed the axiom audit, and was caught by an independent reviewer.

FINITE_DOMAIN_STABLE, IN OUR ADMISSIBLE FILE

Assumed: for every admissible mode n, NOT(det(J - qD) < 0).
Concluded: for every n, det(J - qD) >= 0.
A restatement. The length hypothesis is unused. It is **not** linear asymptotic stability.

A clean axiom audit certifies proof dependencies, not whether the statement says what its name claims. The reviewer's check did.

Replacement: a revised module compiles *every admissible mode has trace < 0 and det > 0* (exit 0, standard axioms). Trace is derived; determinant positivity is assumed. No new mutants yet; eigenvalue and PDE correspondence not established.

This is the system working. Caught by independent review with exact witnesses, not by a machine detecting autonomously.

FIVE MORE VALIDITY ERRORS THIS TEAM MADE

ERROR	CAUGHT BY	LEDGER
1. Thompson: pooled regression read against a buckling envelope	independent critic review	P05 (pending)
2. A diffusion-ratio threshold generalized beyond the model it came from	independent review; Lean-checked classification and exact determinant	F04
3. Pinned liquid-bridge key used an inadmissible volume-changing mode	independent Codex API review (not blinded)	F05
4. Benchmark requests left out declared tolerances and conventions	independent Codex API review	F08
5. Turing scorecard line missing the requirement that det(J - qD) < 0 at an actual admissible q	independent Codex API review	P01 (pending freeze)

The scorecard of 21 candidate laws checked as theorems: every result is pending.

SCORECARD

records in the frozen list	**21**
compiled with proof, axiom audit and a failing mutant	PENDING
excluded, with reasons	PENDING
source-correspondence review	PENDING

The first freeze is not approved; a revised freeze with full formal statements is required before any count is reported.

DO AGENTS NEED THE GATE?

benchmark freeze	PENDING
agent arms and matched budgets	PENDING
coverage, false acceptance, false refusal	PENDING

To count: an always-refuse baseline, the same exact checker without a model, and the unguarded agent. A refusal is not a correct answer. A null will be shown.

MISPREDICTION MAP

Among parameter sets that pass the textbook Turing test, the fraction that are stable on a finite domain of length L, over a large ensemble: PENDING Not yet run on the GPUs; earlier toy-ensemble figures are not shown.

No scores are shown. An earlier benchmark version was retired for input errors. Its successor's numerical verdicts reproduced, but its freeze was withheld because answer information leaks through public IDs. We do not claim that more agents beat fewer.

What this is, what it is not, and whose work it stands on.

CREDIT

Collaborative formalization architecture: **Prove2Me** (arXiv 2608.28433). Prior formal biology: **ATLAS** (facebookresearch/atlas-lean, MathlibExt/Biology), **Rashid et al. 2017** (HOL Light), **lean-bio**, **crnt-lean**, **Ripple** (arXiv 2607.13531). We aim to complement ATLAS, not compete with it. Price–Fisher–Hamilton: Price, Frank. Turing: Turing, Murray.

WE DO NOT CLAIM

Any experiment, a first library, a universal threshold, that any named law is wrong (a scope condition refers to the extracted record), or that agents beat a plain check.

STILL PENDING

The 21-record scorecard · the agent evaluation · Fisher additive-genetic · second-reader review of the Price statements · mutants for the replacement theorem · Turing statements offered to ATLAS · public use of the tool.

Preserved baseline. The earlier enzyme work stays in the appendix with its nulls: 159,129 measured variants; specialists 78/100 equal the single planner 78/100; a plain lazy search reaches 96/100.

Make the assumptions behind a biological-model decision executable and auditable.